THE FUNDAMENTAL GROUP OF NUMERICAL GODEAUX SURFACES

JIHAO LIU, WENFEI LIU, AND LEI ZHANG

Abstract. We prove that the fundamental group of any numerical Godeaux surface is finite. Hence it is cyclic of order at most five and isomorphic to the torsion subgroup of the Picard group. In particular, a numerical Godeaux surface is simply connected if and only if its Picard group is torsion-free.

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A *numerical Godeaux surface* is a smooth minimal projective surface S of general type with $K_S^2 = 1$ and $p_g(S) = q(S) = 0$. The surfaces are named after Godeaux [BCP11, Introduction], who considered the quotient of a quintic surface in $\mathbb{P}^3$ by a free action of $\mathbb{Z}/5$ [God31]. The torsion subgroup of $\mathrm{Pic}(S)$ is cyclic of order at most 5 [Miy76, Lemma 11 and the Remark after Theorem 2']. See also [Rei78, Theorem 2.1]. The profinite completion of $\pi_1(S)$, that is, the inverse limit of the finite quotients of $\pi_1(S)$ [BCP11, Question 4], is cyclic of order at most 5 [BCP11, Theorem 3(1)].

[FR18, Introduction] attributes to Reid the conjecture that the fundamental group of any numerical Godeaux surface is cyclic of order at most 5, and that for any $1 \leq m \leq 5$, the numerical Godeaux surfaces with fundamental group $\mathbb{Z}/m$ form an irreducible and (close to) rational moduli space. See also [BCP11, Conjecture 1] and [Rei, Section 0.2.1], where Reid conjectures that the numerical Godeaux surfaces whose fundamental group has profinite completion $\mathbb{Z}/2$, and those whose fundamental group has trivial profinite completion, form irreducible moduli spaces. [FR18, Introduction] also points out that a numerical Godeaux surface whose fundamental group is infinite and has trivial profinite completion could not be excluded.

The following results are known, where m denotes the order of the torsion subgroup of $\mathrm{Pic}(S)$.

(1) When $m \in \{4, 5\}$, [CU18, Introduction] and [DR26, Introduction] say that $\pi_1(S) \cong \mathbb{Z}/m$ follows from [Rei78]. See [Rei78, Theorem 2.2] for $m = 4$ and [Rei78, Theorem 1.1] for $m = 5$.
(2) When $m = 3$, we have $\pi_1(S) \cong \mathbb{Z}/3$ [CU18, Theorem 1.1].
(3) When $m = 2$, [CD89, Introduction] states as a conjecture the irreducibility of the moduli space of surfaces with $K^2 = 2$, $p_g = 1$, $q = 0$ and trivial torsion, which contains the étale double covers of these surfaces, and lists $\pi_1(S) \cong \mathbb{Z}/2$ as a consequence; [DR26, Corollary 7] states that $\pi_1(S) \cong \mathbb{Z}/2$. The companion paper [LLZ26, Theorem 15.1] proves that the moduli space of these surfaces is irreducible of dimension 8 and that $\pi_1(S) \cong \mathbb{Z}/2$, by solving the equations of [CCO94, Section 7] order by order, as proposed in [CCO94, pp. 83–84].

Date: November 11, 2026.

2020 *Mathematics Subject Classification.* 14J29, 14F35, 32J25.

Key words and phrases. numerical Godeaux surface, fundamental group, L^2 holomorphic section, von Neumann dimension.

Jihao Liu was partially supported by the National Key R&D Program of China (No. 2024YFA1014400).

Wenfei Liu was partially supported by the National Natural Science Foundation of China (No. 12571046).

Lei Zhang was partially supported by the CAS Project for Young Scientists in Basic Research (No. YSBR-032) and the National Natural Science Foundation of China (Nos. 12122116 and 12471495).

(4) When $m = 1$, the Barlow surface is simply connected [Bar85, Theorem 2.5 and the Corollary after it], and so is the Craighero–Gattazzo surface [CG94] by [RTU17, Theorem 6.2].

[BCP11, Question 5] asks for the best possible positive number a such that any minimal surface S of general type with $p_g(S) = 0$ and $K_S^2 \leq a$ has finite fundamental group. [BCP11, Question 10] asks, in particular, whether the fundamental group of any minimal surface S of general type with $q(S) = p_g(S) = 0$ and $K_S^2 \leq 3$ is finite.

Main result. We prove the following theorem. The proof does not use the results (1)–(4) above.

Theorem 1.1. *Let S be a numerical Godeaux surface. Then $\pi_1(S)$ is finite. More precisely, $\pi_1(S)$ is a cyclic group of order at most 5, and $\pi_1(S)$ is isomorphic to the torsion subgroup of $\mathrm{Pic}(S)$. In particular, S is simply connected if and only if the torsion subgroup of $\mathrm{Pic}(S)$ is trivial.*

Theorem 1.1 proves the part of Reid's conjecture in [FR18, Introduction] on the fundamental group. The irreducibility part, in the form of [BCP11, Conjecture 1], is proved in the companion paper [LLZ26, Theorem 1.4], which also proves the irreducibility conjectured by Reid in [Rei, Section 0.2.1]. When the torsion subgroup of $\mathrm{Pic}(S)$ is $\mathbb{Z}/2$, Theorem 1.1 gives $\pi_1(S) \cong \mathbb{Z}/2$, which [CD89, Introduction] lists as a consequence of their conjecture. Since a minimal surface S of general type with $p_g(S) = 0$ satisfies $q(S) = 0$ and $K_S^2 \geq 1$ [BCP11, p. 6], Theorem 1.1 also shows that $a \geq 1$ in [BCP11, Question 5], and answers [BCP11, Question 10] affirmatively when $K_S^2 = 1$. Since a finite group G is isomorphic to the profinite completion of G, Theorem 1.1 also shows that a numerical Godeaux surface whose fundamental group has trivial profinite completion is simply connected.

In the companion paper [LLZ26, Theorem 16.3], we prove that every numerical Godeaux surface with torsion-free Picard group is deformation equivalent to the Barlow surface, and hence simply connected, as part of the classification of numerical Godeaux surfaces. The proof here uses L^2 holomorphic forms on the universal cover.

Sketch of the proof. Assume that $\Gamma := \pi_1(S)$ is infinite. Let $p\colon \widetilde{S} \to S$ be the universal cover, and let B be the Hilbert space of L^2 holomorphic 2-forms on $\widetilde{S}$ (Set-up 2.3), with the unitary action of Γ. Since the profinite completion of Γ is finite, the Γ-reduction of [Cam94] implies that no compact connected analytic subset of $\widetilde{S}$ of positive dimension contains a very general point of $\widetilde{S}$ (Definition 2.4(4) and Lemma 3.1). By the L^2 index theorem [Kol95, Theorem 6.3], the von Neumann dimension of B (Definition 2.7) is at least 1 (Lemma 4.1).

Next, for any non-zero closed Γ-invariant subspace $H \subset B$, the elements of H have no common zero (Proposition 3.7). Hence for any $x \in \widetilde{S}$, the evaluation map at the points of the orbit Γx is an injective bounded operator $B \to \ell^2(\Gamma)$, and since the von Neumann dimension of B is at least 1, the image is dense (Lemma 4.3). Thus the Poincaré series [Kol95, Definition 5.10] of the square of some element of B descends to a member of $H^0(S, 2K_S)$ which does not vanish at $p(x)$. Hence $|2K_S|$ has no base point, which is not possible since $h^0(S, 2K_S) = 2$ and $(2K_S)^2 = 4$. Therefore Γ is finite, and Theorem 1.1 follows from [BCP11, Theorem 3(1)] and Lemma 2.1(3).

The key step is Proposition 3.7. The ideal sheaf generated by the elements of H descends to an ideal sheaf $I \subset \mathcal{O}_S$ (Construction-Proposition 3.3). On a log resolution $\mu\colon T \to S$ of I with $I \cdot \mathcal{O}_T = \mathcal{O}_T(-D)$ [Laz04, Definition 9.1.12], an orthonormal basis of H defines, after division by the canonical section of $\mathcal{O}_T(D)$, a smooth Hermitian metric on $M := \mu^* K_S - D$ whose Chern form [Voi02, Section 3.3.1] is semi-positive (Construction-Proposition 3.5), so M is nef. By Lemma 3.1 and a compactness argument of [Gro91], the differentials of the ratios of elements of H span the cotangent space at some point (Lemma 3.6), so the Chern form is positive definite on a non-empty open subset, and M^2 is a positive integer. Since $K_S^2 = 1$, we have

$$1 \leq M^2 \leq M \cdot \mu^* K_S = 1 - \mu^* K_S \cdot D \leq 1,$$

and the Hodge index theorem implies $D = 0$. We use the L^2 methods of [Gro91] and [Kol95] throughout the paper.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce the von Neumann dimension of closed invariant spaces of L^2 sections on the universal cover, and recall the L^2 index theorem and the Γ-reduction. In Section 3 we prove that, on the universal cover of a numerical Godeaux surface with infinite fundamental group, the elements of any non-zero closed invariant subspace of L^2 holomorphic 2-forms have no common zero. In Section 4 we prove Theorem 1.1.

Acknowledgements. The authors would like to thank Fabrizio Catanese for helpful comments, in particular for pointing out [CD89] and [CCO94]. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance (the Pharos system). Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

Disclosure of AI use. AI tools were used in developing mathematical arguments, checking computations, and drafting and revising this manuscript. The authors are responsible for the mathematical claims, proofs, computations, and references in the final version.

2. Preliminaries

In this section, we recall the facts on numerical Godeaux surfaces, on L^2 holomorphic sections on universal covers, and on the Γ-reduction that we use.

2.1. Numerical Godeaux surfaces.

Lemma 2.1. *Let S be a numerical Godeaux surface. Then the following hold.*

(1) $\chi(\mathcal{O}_S) = 1$, and K_S is nef and big.
(2) $h^0(S, 2K_S) = 2$, and $|2K_S|$ has a base point.
(3) $c_1\colon \mathrm{Pic}(S) \to H^2(S, \mathbb{Z})$ is an isomorphism, and the torsion subgroup of $\mathrm{Pic}(S)$ is isomorphic to the torsion subgroup of $H_1(S, \mathbb{Z})$.

Proof. Since $p_g(S) = q(S) = 0$, we have $\chi(\mathcal{O}_S) = 1 - q(S) + p_g(S) = 1$. Moreover, K_S is nef [BHPV04, Chapter VII, Corollary 2.3], and K_S is big since $K_S^2 = 1 > 0$ [KM98, Proposition 2.61]. This is (1).

By (1), $H^i(S, -K_S) = 0$ for any $i < 2$ [KM98, Theorem 2.70], so $H^i(S, 2K_S) = 0$ for any $i > 0$ by Serre duality [Har77, Corollary III.7.7]. The Riemann–Roch theorem [Har77, Theorem V.1.6] implies that

$$h^0(S, 2K_S) = \chi(\mathcal{O}_S) + \frac{1}{2}(2K_S) \cdot (2K_S - K_S) = 2.$$

If $|2K_S|$ had no base point, then a basis of $H^0(S, 2K_S)$ would define a morphism $f\colon S \to \mathbb{P}^1$ with $f^*\mathcal{O}_{\mathbb{P}^1}(1) \cong \mathcal{O}_S(2K_S)$, and $(2K_S)^2 = 0$ since two distinct fibers of f are disjoint. This is not possible since $(2K_S)^2 = 4$. This is (2).

The exponential sequence [Har77, Appendix B, Section 5] with GAGA [Har77, Appendix B, Theorem 2.1] and $H^1(S, \mathcal{O}_S) = H^2(S, \mathcal{O}_S) = 0$ implies that c_1 is an isomorphism. By the universal coefficient theorem [Hat02, Theorem 3.2], the torsion subgroup of $H^2(S, \mathbb{Z})$ is isomorphic to $\mathrm{Ext}(H_1(S, \mathbb{Z}), \mathbb{Z})$, which is isomorphic to the torsion subgroup of $H_1(S, \mathbb{Z})$ since $H_1(S, \mathbb{Z})$ is a finitely generated abelian group. This is (3). □

Theorem 2.2 ([BCP11, Theorem 3(1)])**.** *Let S be a numerical Godeaux surface. Then the profinite completion of $\pi_1(S)$ is a cyclic group of order at most 5.*

2.2. L^2 holomorphic sections.

Set-up 2.3. Let S, p, Γ, ω_S, dV, and B be as follows.

(1) S is a smooth projective surface.
(2) $p\colon \widetilde{S} \to S$ is the universal cover, on which $\Gamma := \pi_1(S)$ acts by deck transformations.
(3) ω_S is a Kähler form on S.
(4) dV is the volume form of the Kähler form $p^*\omega_S$ on $\widetilde{S}$.
(5) For a complex vector bundle E on S with a Hermitian metric, p^*E carries the pullback metric.
(6) For a section f of p^*E and $\gamma \in \Gamma$, $\gamma^* f$ is the section with $(\gamma^* f)(y) = f(\gamma y)$, where the fibers of p^*E at γy and at y are both the fiber $E_{p(y)}$ of E at $p(y)$.
(7) $L^2(\widetilde{S}, p^*E)$ is the Hilbert space of measurable sections f of p^*E with $\|f\| < +\infty$, where $\|f\|^2 := \int_{\widetilde{S}} |f|^2 dV$.
(8) For a holomorphic vector bundle E on S with a Hermitian metric, $H^0_{(2)}(\widetilde{S}, p^*E) \subset L^2(\widetilde{S}, p^*E)$ is the subspace of holomorphic sections.
(9) K_S carries the Hermitian metric induced by ω_S.
(10) $B := H^0_{(2)}(\widetilde{S}, K_{\widetilde{S}})$, where $K_{\widetilde{S}} = p^*K_S$.

The operators γ^* are unitary on $L^2(\widetilde{S}, p^*E)$, and $(\gamma\delta)^* = \delta^*\gamma^*$ for any $\gamma, \delta \in \Gamma$.

Definition 2.4. Notation as in Set-up 2.3.

(1) A subspace $V \subset L^2(\widetilde{S}, p^*E)$ is Γ-*invariant* if $\gamma^* V = V$ for any $\gamma \in \Gamma$.
(2) An open subset $\widetilde{S}_0 \subset \widetilde{S}$ is an *open fundamental domain* if the subsets $\gamma\widetilde{S}_0$ for $\gamma \in \Gamma$ are pairwise disjoint and the complement of their union in $\widetilde{S}$ has measure zero (cf. [Kol95, (5.6.2) and p. 79]).
(3) Let Y be a complex manifold. A function u on Y *tends to zero at infinity* if for any real number $\epsilon > 0$, there exists a compact subset $K \subset Y$ with $|u| < \epsilon$ on $Y \setminus K$.
(4) Let Y be a connected complex manifold. A property holds for a *very general* point of Y if it holds for any point of Y outside a countable union of proper closed analytic subsets of Y.

Since a countable union of proper closed analytic subsets of a connected complex manifold Y has empty interior, the very general points of Y are dense in Y.

Lemma 2.5 (cf. [Kol95, p. 79])**.** *There exists an open fundamental domain $\widetilde{S}_0 \subset \widetilde{S}$ whose boundary has measure zero.*

Proof. Since S is compact, there exist finitely many open subsets $U_1, \dots, U_N$ covering S, such that each U_k is contained in an open subset $W_k \subset S$ with holomorphic coordinates which map W_k biholomorphically onto the ball $\mathbb{B}(0,2) \subset \mathbb{C}^2$ of radius 2 and U_k onto the ball $\mathbb{B}(0,1)$ of radius 1. Let $A_k := U_k \setminus \bigcup_{l<k} \overline{U_l}$, and let $\widetilde{U}_k$ be a connected component of $p^{-1}(U_k)$. Since U_k is simply connected, $\widetilde{U}_k$ maps isomorphically onto U_k, and the connected components of $p^{-1}(U_k)$ are the subsets $\gamma\widetilde{U}_k$ for $\gamma \in \Gamma$, which are pairwise disjoint. Then the subset

$$\widetilde{S}_0 := \bigcup_{k=1}^{N} \left(\widetilde{U}_k \cap p^{-1}(A_k)\right)$$

is open, and p maps $\widetilde{S}_0$ bijectively onto $\bigcup_k A_k$, so the subsets $\gamma\widetilde{S}_0$ for $\gamma \in \Gamma$ are pairwise disjoint. For any $s \in S$ outside $\bigcup_k \partial U_k$, the smallest k with $s \in \overline{U_k}$ satisfies $s \in A_k$, and $p^{-1}(A_k)$ is the union of the subsets $\gamma(\widetilde{U}_k \cap p^{-1}(A_k))$ for $\gamma \in \Gamma$. Thus the complement of $\bigcup_\gamma \gamma\widetilde{S}_0$ is contained in $p^{-1}(\bigcup_k \partial U_k)$, which has measure zero. Let z be a point of the boundary of $\widetilde{S}_0$, and suppose that $p(z) \in A_k$ for some k. Then z belongs to the open subset $O := \gamma(\widetilde{U}_k \cap p^{-1}(A_k))$ for some

$\gamma \in \Gamma$. Since z belongs to the closure of $\widetilde{S}_0$, the open subset O contains a point $w \in \widetilde{S}_0$. Since $p(w) \in A_k$, we have $w \in \widetilde{U}_k \cap \gamma\widetilde{U}_k$, so $\gamma = 1$, and $z \in \widetilde{S}_0$. This is not possible since $\widetilde{S}_0$ is open. Thus the boundary of $\widetilde{S}_0$ is contained in $p^{-1}(\bigcup_k \partial U_k)$ as well. □

We fix an open fundamental domain $\widetilde{S}_0$ as in Lemma 2.5.

Lemma 2.6. *Notation as in Set-up 2.3. Let L be a holomorphic line bundle on S with a Hermitian metric. Then there exists a real number $C > 0$ such that the following hold.*

(1) For any $f \in H^0_{(2)}(\widetilde{S}, p^*L)$ and any $y \in \widetilde{S}$, we have

$$\sum_{\gamma\in\Gamma} |f(\gamma y)|^2 \le C\|f\|^2.$$

In particular, $|f| \le C^{1/2}\|f\|$ on $\widetilde{S}$.

(2) For any $f \in H^0_{(2)}(\widetilde{S}, p^*L)$, the function $|f|$ tends to zero at infinity.

(3) Let $U \subset S$ be a simply connected open subset with holomorphic coordinates z_1, z_2, let $\widetilde{U}$ be a connected component of $p^{-1}(U)$, and let e be a nowhere vanishing section of L over U. Then for any compact subset $K \subset U$, there exists a real number $C_K > 0$ such that for any $f \in H^0_{(2)}(\widetilde{S}, p^*L)$, the holomorphic function $u := f/p^*e$ on $\widetilde{U}$ satisfies

$$\Big|\frac{\partial u}{\partial(z_k \circ p)}\Big| \le C_K\|f\| \quad \text{on } \widetilde{U} \cap p^{-1}(K)$$

for $k = 1, 2$.

Proof. Since S is compact, there exist finitely many open subsets $U_1, \dots, U_N \subset S$, holomorphic coordinates on each U_k which map U_k biholomorphically onto the ball $\mathbb{B}(0,3) \subset \mathbb{C}^2$ of radius 3, and nowhere vanishing sections e_k of L over U_k, such that the subsets $U'_k \subset U_k$ corresponding to $\mathbb{B}(0,1)$ cover S. Let dV_S be the volume form of ω_S, so that $dV = p^*dV_S$. There exists a real number $c \ge 1$ such that on the subset of each U_k corresponding to the closed ball $\overline{\mathbb{B}}(0,2)$, we have $c^{-1} \le |e_k|^2 \le c$ and $c^{-1}d\lambda \le dV_S \le c\,d\lambda$, where $d\lambda$ is the Lebesgue measure in the coordinates of U_k.

Let $y \in \widetilde{S}$, and fix k with $p(y) \in U'_k$. Since U_k is simply connected, the connected component W_y of $p^{-1}(U_k)$ containing y maps isomorphically onto U_k. Let $W'_y \subset W_y$ be the subset corresponding to the ball of radius 1 centered at $p(y)$, and write $f = g \cdot p^*e_k$ on W_y. Since g is holomorphic, $g(y)$ is the average of g on the ball of radius 1 centered at $p(y)$ with respect to $d\lambda$, so the Cauchy–Schwarz inequality implies

$$|f(y)|^2 \le c|g(y)|^2 \le \frac{c}{\operatorname{vol}\mathbb{B}(0,1)}\int_{W'_y}|g|^2 d\lambda \le \frac{c^3}{\operatorname{vol}\mathbb{B}(0,1)}\int_{W'_y}|f|^2 dV.$$

Since $p(\gamma y) = p(y)$ for any $\gamma \in \Gamma$, we may use k for γy as well, and then $W'_{\gamma y} = \gamma W'_y$. Since the subsets γW_y for $\gamma \in \Gamma$ are pairwise disjoint, summing over γ implies (1) with $C := c^3/\operatorname{vol}\mathbb{B}(0,1)$.

For (2), the segments in the closed ball $\overline{\mathbb{B}}(0,2)$ in the coordinates of each U_k have bounded lengths with respect to ω_S, so there exists a real number $R > 0$ such that W'_y is contained in the ball of radius R centered at y for the distance of $p^*\omega_S$, for any y. Let $\epsilon > 0$. Since $f \in L^2(\widetilde{S}, p^*L)$, there exists a compact subset $K_0 \subset \widetilde{S}$ with

$$\int_{\widetilde{S}\setminus K_0}|f|^2 dV < \frac{\epsilon^2}{C}.$$

Since the metric $p^*\omega_S$ is complete, the set K of points of distance at most R from K_0 is compact. If $y \notin K$, then $W'_y \cap K_0 = \emptyset$, so

$$|f(y)|^2 \le C\int_{\widetilde{S}\setminus K_0}|f|^2 dV < \epsilon^2.$$

This is (2).

For (3), let $r > 0$ be a real number such that the set K_r of points of U whose coordinates differ from the coordinates of a point of K by at most r in each coordinate is a compact subset of U, and let $c_K := \max_{K_r} |e|^{-1}$. By (1), $|u| \leq c_K C^{1/2} \|f\|$ on $\widetilde{U} \cap p^{-1}(K_r)$, so the Cauchy inequalities on polydiscs of radius r imply (3) with $C_K := c_K C^{1/2}/r$. □

By Lemma 2.6(1), a sequence in $H^0_{(2)}(\widetilde{S}, p^*L)$ which converges in $L^2(\widetilde{S}, p^*L)$ converges uniformly on $\widetilde{S}$. Thus $H^0_{(2)}(\widetilde{S}, p^*L)$ is a closed Γ-invariant subspace of $L^2(\widetilde{S}, p^*L)$.

Definition 2.7 ([Kol95, (6.10.2) and (6.10.3)])**.** Notation as in Set-up 2.3. Let E be a complex vector bundle on S with a Hermitian metric, and let $V \subset L^2(\widetilde{S}, p^*E)$ be a closed Γ-invariant subspace with an orthonormal basis $(s_i)_{i \in I}$. The *von Neumann dimension* of V is

$$\dim_\Gamma V := \sum_{i \in I} \int_{\widetilde{S}_0} |s_i|^2 dV \in [0, +\infty].$$

It does not depend on the choice of the orthonormal basis.

Lemma 2.8 ([Kol95, Lemmas 6.10.4 and 6.11])**.** *Notation as in Definition 2.7. Then the following hold.*

(1) Let $V \subset V'$ be closed Γ-invariant subspaces of $L^2(\widetilde{S}, p^*E)$. Then $\dim_\Gamma V \leq \dim_\Gamma V'$, and if $\dim_\Gamma V = \dim_\Gamma V' < +\infty$, then $V = V'$.

(2) Let Ψ be a bounded linear operator on $L^2(\widetilde{S}, p^*E)$ with $\Psi\gamma^* = \gamma^*\Psi$ for any $\gamma \in \Gamma$. Then

$$\dim_\Gamma (\ker \Psi)^\perp = \dim_\Gamma \overline{\Psi(L^2(\widetilde{S}, p^*E))}.$$

Theorem 2.9 ([Kol95, Definition 6.2.2, p. 73, and Theorem 6.3])**.** *Notation as in Set-up 2.3. For $0 \leq i \leq 2$, let $\Lambda^{2,i}_S$ be the bundle of $(2,i)$-forms on S with the metric induced by ω_S, and let $\mathcal{H}^{2,i}_{(2)}(\widetilde{S})$ be the closed Γ-invariant subspace of $L^2(\widetilde{S}, p^*\Lambda^{2,i}_S)$ of forms α with $\Delta_{\bar\partial}\alpha = 0$, where $\Delta_{\bar\partial}$ is the Laplacian of $\bar\partial$ with respect to $p^*\omega_S$. Then*

$$\chi(S, K_S) = \sum_{i=0}^{2} (-1)^i \dim_\Gamma \mathcal{H}^{2,i}_{(2)}(\widetilde{S}).$$

Moreover, $\mathcal{H}^{2,0}_{(2)}(\widetilde{S}) = B$, and the space $\mathcal{H}^{0,0}_{(2)}(\widetilde{S})$ of L^2 functions u on $\widetilde{S}$ with $\Delta_{\bar\partial} u = 0$ is $H^0_{(2)}(\widetilde{S}, \mathcal{O}_{\widetilde{S}})$.

2.3. **The Γ-reduction.**

Construction-Theorem 2.10 ([Cam94, Théorème 3.5, Corollaire 3.7, Définition 3.8, and p. 269])**.** Notation as in Set-up 2.3(1)(2). Then there exist an analytic space $\widetilde{Z}$ with an action of Γ and a Γ-equivariant proper surjective meromorphic map $\tilde\rho$ from $\widetilde{S}$ to $\widetilde{Z}$ with connected fibers, which induces a proper surjective meromorphic map ρ from S to the compact complex analytic space $Z := \widetilde{Z}/\Gamma$ with connected fibers, such that the following hold.

(1) $\rho \circ p = q \circ \tilde\rho$, where $q\colon \widetilde{Z} \to Z$ is the quotient map.

(2) For a very general point $y \in \widetilde{S}$, the fiber $\tilde\rho^{-1}(\tilde\rho(y))$ is the largest compact connected analytic subset of $\widetilde{S}$ containing y.

(3) If $\dim Z = 1$, then there exist a compact Riemann surface C of genus at least one, subgroups of finite index $\Gamma' \subset \Gamma$ and $\Pi \subset \pi_1(C)$, and finite normal subgroups $F \subset \Gamma'$ and $F' \subset \Pi$ with $\Gamma'/F \cong \Pi/F'$.

Here a meromorphic map $\tilde\rho\colon \widetilde{S} \dashrightarrow \widetilde{Z}$ is given by the graph $G \subset \widetilde{S} \times \widetilde{Z}$, which is an irreducible analytic subset such that the projection $G \to \widetilde{S}$ is a proper modification [Cam94, Section 3.2]. The map $\tilde\rho$ is proper and surjective if the projection $G \to \widetilde{Z}$ is proper and surjective, and the

fiber $\tilde{\rho}^{-1}(z)$ of $\tilde{\rho}$ over $z \in \widetilde{Z}$ is the image of $G \cap (\widetilde{S} \times \{z\})$ in $\widetilde{S}$. The map ρ is the Γ-*reduction* of S [Cam94, Corollaire 3.7 and Définition 3.8]. Item (3) is the statement of [Cam94, p. 269] that Γ is commensurable with $\pi_1(C)$ when the Γ-dimension $\dim Z$ [Cam94, Définition 4.1] is one. Commensurability is not defined in [Cam94], and (3) holds both if commensurable groups are groups with isomorphic subgroups of finite index (F and F' trivial) and if commensurability is understood up to finite normal subgroups. In [Cam94, Théorème 3.5], a general point is a very general point in the sense of Definition 2.4(4) [Cam94, Définition 3.3].

3. Invariant subspaces of L^2 holomorphic 2-forms

In this section, we prove Proposition 3.7: on the universal cover of a numerical Godeaux surface with infinite fundamental group, the elements of any non-zero closed Γ-invariant subspace of B have no common zero.

Lemma 3.1. *Notation and conditions as in Set-up 2.3. Assume that S is a numerical Godeaux surface and Γ is infinite. Then there exists a countable union Ξ of proper closed analytic subsets of $\widetilde{S}$ such that for any $y \in \widetilde{S} \setminus \Xi$, any compact connected analytic subset of $\widetilde{S}$ containing y is $\{y\}$.*

Proof. Let $\tilde{\rho}$, ρ, $\widetilde{Z}$, Z, and q be as in Construction-Theorem 2.10. Since ρ is surjective, $\dim Z \leq \dim S = 2$.

Suppose that $\dim Z = 0$. Since S is connected and ρ is surjective, Z is a point, so $\widetilde{Z} = q^{-1}(Z)$ is a single Γ-orbit. Since the graph of $\tilde{\rho}$ is irreducible and the projection from the graph to $\widetilde{Z}$ is surjective, $\widetilde{Z}$ is connected. Since Γ is countable, $\widetilde{Z}$ is a point. By Construction-Theorem 2.10(2), $\widetilde{S} = \tilde{\rho}^{-1}(\widetilde{Z})$ is compact, which is not possible since Γ is infinite.

Suppose that $\dim Z = 1$. By Construction-Theorem 2.10(3), there exist a compact Riemann surface C of genus at least one, subgroups of finite index $\Gamma' \subset \Gamma$ and $\Pi \subset \pi_1(C)$, and finite normal subgroups $F \subset \Gamma'$ and $F' \subset \Pi$, with $\Gamma'/F \cong \Pi/F'$. Since $\pi_1(C)$ is torsion-free, F' is trivial, so there exists a surjective homomorphism $\Gamma' \to \Pi$. We have $\Pi \cong \pi_1(C')$ for a finite étale cover $C' \to C$, and C' has genus at least one, so there exists a surjective homomorphism $\Pi \to H_1(C', \mathbb{Z}) \to \mathbb{Z}$. Thus for any positive integer n, there exists a surjective homomorphism $\Gamma' \to \mathbb{Z}/n$. Let Γ_n be the kernel of such a homomorphism. Then $[\Gamma : \Gamma_n] = n[\Gamma : \Gamma']$, and the intersection N_n of the conjugates of Γ_n in Γ is a normal subgroup of finite index of Γ with $[\Gamma : N_n] \geq n$. Hence Γ has finite quotients of arbitrarily large order, so the profinite completion of Γ is infinite, which contradicts Theorem 2.2.

Thus $\dim Z = 2$. Since ρ is proper and surjective with connected fibers and $\dim Z = \dim S$, the fibers of the projection from the graph of ρ to Z are points outside a proper closed analytic subset of Z. Thus there exists a proper closed analytic subset $Y_\rho \subset S$ such that for any $s \in S \setminus Y_\rho$, the map ρ is holomorphic near s and $\rho^{-1}(\rho(s)) = \{s\}$. Let $\Xi \subset \widetilde{S}$ be the union of $p^{-1}(Y_\rho)$, the indeterminacy locus of $\tilde{\rho}$, and the countable union of proper closed analytic subsets of $\widetilde{S}$ outside which Construction-Theorem 2.10(2) holds. Let $y \in \widetilde{S} \setminus \Xi$, and let $W \subset \widetilde{S}$ be a compact connected analytic subset containing y. Then $W \subset \tilde{\rho}^{-1}(\tilde{\rho}(y))$. By Construction-Theorem 2.10(1), the map $p \times q$ sends the graph of $\tilde{\rho}$ into the graph of ρ, so

$$p(\tilde{\rho}^{-1}(\tilde{\rho}(y))) \subset \rho^{-1}(q(\tilde{\rho}(y))) = \rho^{-1}(\rho(p(y))) = \{p(y)\}.$$

Thus $W \subset p^{-1}(p(y))$, which is a discrete subset of $\widetilde{S}$. Since W is connected, $W = \{y\}$. □

Set-up 3.2. Notation and conditions as in Set-up 2.3. Let H, $(h_j)_{j \in J}$, r_H, and $r(H)$ be as follows.

(1) S is a numerical Godeaux surface.
(2) Γ is infinite.
(3) $H \subset B$ is a non-zero closed Γ-invariant subspace.
(4) $(h_j)_{j \in J}$ is an orthonormal basis of H, where J is a countable set.

(5) For $y \in \widetilde{S}$ and $h_0 \in H$ with $h_0(y) \neq 0$, $r_H(y)$ is the dimension of the subspace of the holomorphic cotangent space $T_y^*\widetilde{S}$ spanned by the differentials $d(h/h_0)(y)$ for $h \in H$.
(6) $r(H)$ is the maximum of $r_H(y)$, where y runs over the points of $\widetilde{S}$ at which some element of H does not vanish.

An orthonormal basis as in (4) exists since $L^2(\widetilde{S}, K_{\widetilde{S}})$ is separable. By the quotient rule, $r_H(y)$ does not depend on the choice of h_0. Since r_H is lower semicontinuous, the set of points $y \in \widetilde{S}$ with $r_H(y) = r(H)$ is open and non-empty.

Construction-Proposition 3.3. Notation as in Set-up 3.2. Let $\mathcal{I} \subset \mathcal{O}_{\widetilde{S}}$ be the ideal sheaf generated by the functions h/e, where $h \in H$ and e runs over the nowhere vanishing local sections of $K_{\widetilde{S}}$. Then the following hold.

(1) $\mathcal{I}$ is coherent.
(2) $\mathcal{I}$ is generated by the functions h_j/e for $j \in J$.
(3) There exists a coherent ideal sheaf $I \subset \mathcal{O}_S$ with $\mathcal{I} = p^{-1}I \cdot \mathcal{O}_{\widetilde{S}}$.
(4) The set of common zeros of the elements of H is $p^{-1}(V(I))$.

Proof. Let $\mathcal{G} \subset K_{\widetilde{S}}$ (resp. $\mathcal{G}' \subset K_{\widetilde{S}}$) be the subsheaf generated by the sections $h \in H$ (resp. h_j for $j \in J$). By [GR84, Chapter 5, §6.1, Corollary of the Noether Lemma], $\mathcal{G}$ and $\mathcal{G}'$ are coherent, and $\mathcal{G}' \subset \mathcal{G}$. Let $h \in H$, and write $h = \sum_j c_j h_j$, where the series converges in $L^2(\widetilde{S}, K_{\widetilde{S}})$. By Lemma 2.6(1), the partial sums of the series, which are sections of $\mathcal{G}'$, converge to h uniformly on $\widetilde{S}$. By the closedness of submodules [GR84, Chapter 2, §2.3], the germ of h at any point of $\widetilde{S}$ belongs to $\mathcal{G}'$. Hence $\mathcal{G} = \mathcal{G}'$, and $\mathcal{I} = \mathcal{G} \otimes K_{\widetilde{S}}^{-1}$. This is (1) and (2).

Since $\gamma^* H = H$, we have $\gamma^*\mathcal{I} = \mathcal{I}$ for any $\gamma \in \Gamma$. For a simply connected open subset $U \subset S$ and a connected component $\widetilde{U}$ of $p^{-1}(U)$, we let $I|_U$ be the ideal sheaf corresponding to $\mathcal{I}|_{\widetilde{U}}$ under the isomorphism $p|_{\widetilde{U}} \colon \widetilde{U} \to U$. Since $\gamma^*\mathcal{I} = \mathcal{I}$, the ideal sheaf $I|_U$ does not depend on $\widetilde{U}$, so the ideal sheaves $I|_U$ glue to a coherent ideal sheaf $I \subset \mathcal{O}_S$ with $\mathcal{I} = p^{-1}I \cdot \mathcal{O}_{\widetilde{S}}$. This is (3). The set of common zeros of the elements of H is the zero set of $\mathcal{I}$, which is $p^{-1}(V(I))$. This is (4). □

By GAGA [Har77, Appendix B, Theorem 2.1], the ideal sheaf I is algebraic.

Construction-Proposition 3.4. Notation as in Construction-Proposition 3.3. Let $\mu \colon T \to S$ be a log resolution of I with $I \cdot \mathcal{O}_T = \mathcal{O}_T(-D)$ [Laz04, Definition 9.1.12 and Example 9.1.14], let s_D be the canonical section of $\mathcal{O}_T(D)$, and let $M := \mu^* K_S - D$. Let $\widetilde{T} := \widetilde{S} \times_S T$, with the projections $p_T \colon \widetilde{T} \to T$ and $\tilde{\mu} \colon \widetilde{T} \to \widetilde{S}$, on which Γ acts by $\gamma(y, t) = (\gamma y, t)$, and define γ^* on sections of $p_T^* M$ as in Set-up 2.3(6). Then the following hold.

(1) For any $h \in H$, there exists a unique holomorphic section φ_h of $p_T^* M$ with $\tilde{\mu}^* h = \varphi_h \cdot p_T^* s_D$.
(2) $\gamma^* \varphi_h = \varphi_{\gamma^* h}$ for any $h \in H$ and $\gamma \in \Gamma$.
(3) The sections φ_h for $h \in H$ have no common zero on $\widetilde{T}$.

Proof. Let $h \in H$, $y \in \widetilde{S}$, and let e be a nowhere vanishing section of $K_{\widetilde{S}}$ near y. By Construction-Proposition 3.3(3), the function h/e is a section of $\mathcal{I} = p^{-1}I \cdot \mathcal{O}_{\widetilde{S}}$ near y, so near any point of $\tilde{\mu}^{-1}(y)$, the section $\tilde{\mu}^* h$ is a section of

$$(\mathcal{I} \cdot \mathcal{O}_{\widetilde{T}}) \otimes \tilde{\mu}^* K_{\widetilde{S}} = p_T^*(\mathcal{O}_T(-D) \otimes \mu^* K_S).$$

Since $p_T^* s_D$ does not vanish identically on any non-empty open subset of $\widetilde{T}$, there exists a unique section φ_h as in (1). By the uniqueness, $\gamma^* \varphi_h = \varphi_{\gamma^* h}$ for any $\gamma \in \Gamma$. This is (2).

Let $t \in \widetilde{T}$ and $y := \tilde{\mu}(t)$, and let e be a nowhere vanishing section of $K_{\widetilde{S}}$ near y. By Construction-Proposition 3.3(2) and the Noether property [GR84, Chapter 5, §6.1, (N)] of the ideal sheaves generated by the functions h_j/e for j in finite subsets of J, there exists a finite subset $F \subset J$ such that $\mathcal{I}$ is generated by the functions h_j/e for $j \in F$ near y. Write $p_T^* s_D = g\sigma$

near t, where g is a holomorphic function and σ is a nowhere vanishing section of $p_T^*\mathcal{O}_T(D)$, and let $\eta := \tilde{\mu}^* e \otimes \sigma^{-1}$, which is a nowhere vanishing section of p_T^*M near t. Then

$$\varphi_{h_j} = g^{-1} \cdot \big((h_j/e) \circ \tilde{\mu}\big) \cdot \eta.$$

Since $I \cdot \mathcal{O}_T = \mathcal{O}_T(-D)$, the functions $(h_j/e) \circ \tilde{\mu}$ for $j \in F$ generate the ideal $g \cdot \mathcal{O}_{\widetilde{T},t}$. Thus the functions φ_{h_j}/η for $j \in F$ generate $\mathcal{O}_{\widetilde{T},t}$, and $\varphi_{h_j}(t) \neq 0$ for some $j \in F$. This is (3). □

Let θ be a real $(1,1)$-form on a 2-dimensional complex manifold, written locally as

$$\theta = \frac{i}{2\pi} \sum_{k,l=1}^{2} \theta_{k\bar{l}}\, dz_k \wedge d\bar{z}_l.$$

Then θ is *semi-positive* (resp. *positive definite*) at a point if the Hermitian matrix $(\theta_{k\bar{l}})$ is positive semi-definite (resp. positive definite) at the point, and it is *semi-positive* if it is semi-positive at any point.

Construction-Proposition 3.5. Notation as in Construction-Proposition 3.4. Let $S^\circ \subset S$ be the complement of a finite subset such that μ is an isomorphism over S° [Har77, Corollary V.5.4], and let $\Omega \subset \widetilde{S}$ be the set of points y with $r_H(y) = 2$. Then there exists a smooth Hermitian metric on M with Chern form θ [Voi02, Section 3.3.1] such that the following hold.

(1) θ is semi-positive.
(2) θ is positive definite on $\mu^{-1}(p(\Omega) \cap S^\circ)$.
(3) For any $h \in H$, we have $|\varphi_h| \leq \|h\|$ on $\widetilde{T}$ with respect to the pullback metric on p_T^*M.
(4) For any $h \in H$, the function $|\varphi_h|$ tends to zero at infinity on $\widetilde{T}$.

Proof. A *chart* is a tuple $(V, r, \alpha_1, \alpha_2, U, e, \widetilde{U})$ with the following properties.

- $V \subset T$ is an open subset with holomorphic coordinates z_1, z_2 which map V biholomorphically onto the polydisc $\{|z_1| < 2r, |z_2| < 2r\}$, where $r > 0$ is a real number.
- $D|_V$ is defined by $g := z_1^{\alpha_1} z_2^{\alpha_2}$, where α_1, α_2 are non-negative integers.
- $U \subset S$ is a simply connected open subset with holomorphic coordinates.
- $\mu(V) \subset U$.
- e is a nowhere vanishing section of K_S over U.
- $\widetilde{U}$ is a connected component of $p^{-1}(U)$.

For a chart, let $\Delta_r \subset V$ be the polydisc of polyradius (r, r) centered at the origin. Since D has simple normal crossing support, the polydiscs Δ_r of the charts cover T. For a chart, let $\widetilde{V} := \widetilde{U} \times_U V \subset \widetilde{T}$. Since $p|_{\widetilde{U}} \colon \widetilde{U} \to U$ and $p_T|_{\widetilde{V}} \colon \widetilde{V} \to V$ are biholomorphic, we regard points and functions on U and V as points and functions on $\widetilde{U}$ and $\widetilde{V}$ through $p|_{\widetilde{U}}$ and $p_T|_{\widetilde{V}}$. Let σ_V be the nowhere vanishing section of $\mathcal{O}_T(D)$ over V with $s_D = g\sigma_V$, and let $\eta_V := \mu^* e \otimes \sigma_V^{-1}$, which is a nowhere vanishing section of M over V. For $h \in H$, we define holomorphic functions u_h on U and a_h on V by

$$h = u_h \cdot p^* e \text{ on } \widetilde{U}, \qquad \varphi_h = a_h \cdot p_T^* \eta_V \text{ on } \widetilde{V}.$$

Then $a_h = (u_h \circ \mu)/g$. For $j \in J$, we write $u_j := u_{h_j}$ and $a_j := a_{h_j}$.

Step 1. In this step, we show that the series $b_U := \sum_j |u_j|^2$ converges uniformly on compact subsets of U, and that b_U is continuous.

For $z \in U$, the linear form $h \to u_h(z)$ on H is bounded by Lemma 2.6(1). By the Riesz representation theorem, the square of the norm of the linear form $h \to u_h(z)$ is $\sum_j |u_j(z)|^2$, so

$$b_U(z) = \sup\{|u_h(z)|^2 \mid h \in H,\ \|h\| \leq 1\}.$$

Let K be a compact convex subset of U in the coordinates of U. By Lemma 2.6(3), there exists a real number $C_K > 0$ such that $|u_h(z) - u_h(z')| \leq 2C_K|z - z'|$ for any $z, z' \in K$ and any $h \in H$ with $\|h\| \leq 1$. Thus for any $z, z' \in K$,

$$|b_U(z)^{1/2} - b_U(z')^{1/2}| \leq 2C_K|z - z'|,$$

and b_U is continuous. By Dini's theorem, the series converges uniformly on compact subsets of U.

Step 2. In this step, we bound a_h by u_h. Let $R := 3r/2$, let $P \subset V$ be the torus defined by $|z_1| = |z_2| = R$, and let $C_V := R^{2-\alpha_1-\alpha_2}(R-r)^{-2}$. We show that for any $h \in H$ and any $z \in \overline{\Delta}_r$,

$$|a_h(z)| \le C_V \max_{w\in P} |u_h(\mu(w))|, \tag{3.1}$$

and that the series $\sum_j |a_j|^2$ converges uniformly on $\overline{\Delta}_r$.

Since $a_h = (u_h \circ \mu)/g$ is holomorphic on V and $|g| = R^{\alpha_1+\alpha_2}$ on P, the Cauchy integral formula

$$a_h(z) = \frac{1}{(2\pi i)^2} \int_{|w_1|=R} \int_{|w_2|=R} \frac{u_h(\mu(w))}{w_1^{\alpha_1} w_2^{\alpha_2} (w_1 - z_1)(w_2 - z_2)}\, dw_1 dw_2$$

implies (3.1). Let $F \subset J$ be a finite subset, and let $c_j \in \mathbb{C}$ for $j \in F$ with $\sum_{j\in F} |c_j|^2 \le 1$. Applying (3.1) to $h = \sum_{j\in F} c_j h_j$, the Cauchy–Schwarz inequality implies

$$\Big|\sum_{j\in F} c_j a_j(z)\Big| \le C_V \max_{w\in P} \Big(\sum_{j\in F} |u_j(\mu(w))|^2\Big)^{1/2}.$$

Taking the supremum over $(c_j)_{j\in F}$, we get

$$\sum_{j\in F} |a_j(z)|^2 \le C_V^2 \max_{w\in P} \sum_{j\in F} |u_j(\mu(w))|^2. \tag{3.2}$$

For a finite subset $J_0 \subset J$, applying (3.2) to the finite subsets $F \subset J \setminus J_0$, we see that $\sum_{j\notin J_0} |a_j|^2$ is at most C_V^2 times the supremum of $\sum_{j\notin J_0} |u_j|^2$ on the compact subset $\mu(P)$ of U. Thus Step 1 implies that the series $\sum_j |a_j|^2$ converges uniformly on $\overline{\Delta}_r$.

Step 3. In this step, we construct the metric on M. By Step 2 and the Cauchy–Schwarz inequality, the series $\sum_j a_j(z)\overline{a_j(w)}$ converges uniformly on $\Delta_r \times \Delta_r$, so the sum of the series is holomorphic in z and anti-holomorphic in w. Thus the function $\beta_V := \sum_j |a_j|^2$ is continuous on $\overline{\Delta}_r$ and real analytic on Δ_r, and the partial derivatives of β_V on Δ_r are computed termwise. For $h = \sum_j c_j h_j \in H$ and $z \in \overline{\Delta}_r$, the linear form $h \to a_h(z)$ on H is bounded by (3.1), so $a_h = \sum_j c_j a_j$ on $\overline{\Delta}_r$. Since the sections φ_h for $h \in H$ have no common zero (Construction-Proposition 3.4(3)), the functions a_j for $j \in J$ have no common zero, so $\beta_V > 0$ on $\overline{\Delta}_r$.

On the dense open subset of Δ_r where $g \ne 0$, Step 1 implies $\beta_V = (b_U \circ \mu)/|g|^2$. Let $(V', r', \alpha_1', \alpha_2', U', e', \widetilde{U}')$ be another chart, with the corresponding g', $\eta_{V'}$, u_h', $b_{U'}$, and $\beta_{V'}$. Let $z \in U \cap U'$, and let $\tilde{z} \in \widetilde{U}$ and $\gamma\tilde{z} \in \widetilde{U}'$ be the points over z, where $\gamma \in \Gamma$. Then $u_h'(z) = (e/e')(z) \cdot u_{\gamma^* h}(z)$ for any $h \in H$. Since γ^* is a unitary automorphism of H, we get $b_{U'} = |e/e'|^2 b_U$ on $U \cap U'$. Since $\eta_V/\eta_{V'} = ((e/e') \circ \mu) \cdot (g/g')$, we get

$$\beta_{V'} = |\eta_V/\eta_{V'}|^2 \beta_V$$

on the dense open subset of $\Delta_r \cap \Delta_{r'}$ where $gg' \ne 0$, hence on $\Delta_r \cap \Delta_{r'}$ by continuity. Thus there exists a smooth Hermitian metric on M with $|\eta_V|^2 = 1/\beta_V$ on Δ_r for any chart, and by continuity, $|\eta_V|^2 = 1/\beta_V$ on $\overline{\Delta}_r$. The Chern form θ of the metric is $\frac{i}{2\pi}\partial\bar\partial \log \beta_V$ on Δ_r [Voi02, Section 3.3.1].

Step 4. In this step, we prove (1) and (2). Let $t \in \Delta_r$ for a chart, and let $v = (v_1, v_2) \in \mathbb{C}^2$, viewed as a tangent vector of type $(1,0)$ at t. Let

$$\partial_v := v_1 \frac{\partial}{\partial z_1} + v_2 \frac{\partial}{\partial z_2},$$

and consider the vectors $\mathbf{a} := (a_j(t))_{j\in J}$ and $\mathbf{a}' := (\partial_v a_j(t))_{j\in J}$ in $\ell^2(J)$, with the inner product $\langle\cdot,\cdot\rangle$ and the norm $\|\cdot\|$ of $\ell^2(J)$. Since the partial derivatives of β_V are computed termwise,

$$\sum_{k,l=1}^{2} \frac{\partial^2 \log \beta_V}{\partial z_k \partial \bar{z}_l}(t)\, v_k \bar{v}_l = \frac{\|\mathbf{a}\|^2\|\mathbf{a}'\|^2 - |\langle \mathbf{a}', \mathbf{a}\rangle|^2}{\|\mathbf{a}\|^4}. \tag{3.3}$$

By the Cauchy–Schwarz inequality, the right-hand side of (3.3) is non-negative, and it is zero if and only if $\mathbf{a}' = \lambda\mathbf{a}$ for some $\lambda \in \mathbb{C}$. Thus θ is semi-positive. This is (1).

For (2), let $t \in \mu^{-1}(p(\Omega) \cap S^\circ)$, and let $y \in \Omega$ with $p(y) = \mu(t)$. Then μ is an isomorphism over a neighborhood of $\mu(t)$. Choose a chart with $t \in \Delta_r$, where $\widetilde{U}$ is the connected component of $p^{-1}(U)$ containing y. Since $r_H(y) = 2$, there exist $h_0, h_1, h_2 \in H$ with $h_0(y) \neq 0$ and $d(h_1/h_0)(y) \wedge d(h_2/h_0)(y) \neq 0$. Since $u_{h_0}(\mu(t)) \neq 0$ and $u_{h_0} \circ \mu = g \cdot a_{h_0}$, we have $g(t) \neq 0$, $a_{h_0}(t) \neq 0$, and $a_h/a_{h_0} = (u_h/u_{h_0}) \circ \mu$ near t for any $h \in H$. Suppose that the right-hand side of (3.3) is zero for some non-zero v. Then $\mathbf{a}' = \lambda\mathbf{a}$ for some $\lambda \in \mathbb{C}$. For $h = \sum_j c_j h_j \in H$, the series $\sum_j c_j a_j$ converges uniformly on Δ_r by Step 2 and the Cauchy–Schwarz inequality, so $\partial_v a_h(t) = \sum_j c_j \partial_v a_j(t) = \lambda a_h(t)$. Thus $\partial_v(a_h/a_{h_0})(t) = 0$ for any $h \in H$. Since μ is an isomorphism near t and u_{h_k}/u_{h_0} corresponds to h_k/h_0 through the biholomorphism $p|_{\widetilde{U}}$, the vector $d\mu_t(v)$ is annihilated by $d(h_1/h_0)(y)$ and $d(h_2/h_0)(y)$. This contradicts $d(h_1/h_0)(y) \wedge d(h_2/h_0)(y) \neq 0$. Thus θ is positive definite at t. This is (2).

Step 5. In this step, we prove (3) and (4). Let $h = \sum_j c_j h_j \in H$. On the subset of $\widetilde{V}$ over Δ_r, the Cauchy–Schwarz inequality implies

$$|\varphi_h|^2 = \frac{|a_h|^2}{\beta_V} = \frac{|\sum_j c_j a_j|^2}{\sum_j |a_j|^2} \leq \|h\|^2.$$

For $\gamma \in \Gamma$ and $\tilde{t} \in \widetilde{T}$, Construction-Proposition 3.4(2) implies

$$|\varphi_h(\gamma\tilde{t})| = |(\gamma^*\varphi_h)(\tilde{t})| = |\varphi_{\gamma^* h}(\tilde{t})|,$$

and $\|\gamma^* h\| = \|h\|$. Since $\widetilde{T}$ is covered by the subsets $\gamma\widetilde{V}$ over Δ_r for the charts and $\gamma \in \Gamma$, we get $|\varphi_h| \leq \|h\|$ on $\widetilde{T}$. This is (3).

Since T is compact, there exist finitely many charts $(V_k, r_k, \alpha_{k,1}, \alpha_{k,2}, U_k, e_k, \widetilde{U}_k)$ for $1 \leq k \leq N$, with the corresponding P_k and C_{V_k}, such that the closed polydiscs $\overline{\Delta}_{r_k} \subset V_k$ cover T. Let $\widetilde{K}_k \subset \widetilde{U}_k$ be the compact subset over $\mu(P_k)$, and let $\widetilde{K}'_k \subset \widetilde{V}_k$ be the compact subset over $\overline{\Delta}_{r_k}$. There exists a real number $c > 0$ such that $\beta_{V_k} \geq c^{-1}$ on $\overline{\Delta}_{r_k}$ and $|e_k|^{-1} \leq c$ on $\mu(P_k)$ for any k. Let $\tilde{t} \in \widetilde{K}'_k$ and $\gamma \in \Gamma$. Since $|\eta_{V_k}|^2 = 1/\beta_{V_k}$ on $\overline{\Delta}_{r_k}$, (3.1) for $\gamma^* h$ in place of h implies

$$|\varphi_h(\gamma\tilde{t})| = \frac{|a_{\gamma^* h}(\tilde{t})|}{\beta_{V_k}(\tilde{t})^{1/2}} \leq c^{1/2} C_{V_k} \max_{w \in P_k} |u_{\gamma^* h}(\mu(w))| \leq c^{3/2} C_{V_k} \max_{\tilde{w} \in \widetilde{K}_k} |h(\gamma\tilde{w})|.$$

Let $\epsilon > 0$. By Lemma 2.6(2), there exists a compact subset $K \subset \widetilde{S}$ with $c^{3/2} C_{V_k} |h| < \epsilon$ on $\widetilde{S} \setminus K$ for any k. Since Γ acts properly discontinuously on $\widetilde{S}$, the set Γ_0 of elements $\gamma \in \Gamma$ with $\gamma\widetilde{K}_k \cap K \neq \emptyset$ for some k is finite. Thus $|\varphi_h| < \epsilon$ outside the compact subset

$$\bigcup_{\gamma \in \Gamma_0} \bigcup_{k=1}^{N} \gamma \widetilde{K}'_k \subset \widetilde{T}.$$

This is (4). □

Lemma 3.6. *Notation as in Set-up 3.2. Then $r(H) = 2$.*

Proof. Suppose that $r(H) = 0$. Let $h_0 \in H$ be a non-zero element. For any $h \in H$, the function h/h_0 has zero differential on the subset of $\widetilde{S}$ where $h_0 \neq 0$, which is connected since it is the complement of a proper closed analytic subset of the connected manifold $\widetilde{S}$. Thus h/h_0 is constant, and $H = \mathbb{C}h_0$. Since γ^* is a unitary automorphism of H, we have $\gamma^* h_0 = \lambda_\gamma h_0$ with

$|\lambda_\gamma| = 1$, so $|h_0(\gamma y)| = |h_0(y)|$ for any $\gamma \in \Gamma$ and $y \in \widetilde{S}$. Since the subsets $\gamma\widetilde{S}_0$ for $\gamma \in \Gamma$ are pairwise disjoint and the complement of their union has measure zero (Definition 2.4(2)),

$$\|h_0\|^2 = \sum_{\gamma\in\Gamma} \int_{\gamma\widetilde{S}_0} |h_0|^2 dV = \sum_{\gamma\in\Gamma} \int_{\widetilde{S}_0} |h_0|^2 dV,$$

which is infinite since Γ is infinite and h_0 does not vanish identically on the non-empty open subset $\widetilde{S}_0$. This is not possible.

Suppose that $r(H) = 1$. Let μ, T, D, M, $\widetilde{T}$, p_T, $\tilde{\mu}$, and φ_h be as in Construction-Proposition 3.4, and let S° be as in Construction-Proposition 3.5. By Lemma 3.1, there exists a countable union Ξ of proper closed analytic subsets of $\widetilde{S}$ such that for any $y \in \widetilde{S} \setminus \Xi$, any compact connected analytic subset of $\widetilde{S}$ containing y is $\{y\}$. Since the set of points $y \in \widetilde{S}$ with $r_H(y) = 1$ and $p(y) \in S^\circ$ is open and non-empty, and $\widetilde{S} \setminus \Xi$ is dense in $\widetilde{S}$, there exists $y_0 \in \widetilde{S} \setminus \Xi$ with $r_H(y_0) = 1$ and $p(y_0) \in S^\circ$. Let $t_0 \in \widetilde{T}$ be the point over y_0. Fix $h_0 \in H$ with $h_0(y_0) \neq 0$, and write $\varphi_0 := \varphi_{h_0}$. Since $\tilde{\mu}^* h_0 = \varphi_0 \cdot p_T^* s_D$, we have $\varphi_0(t_0) \neq 0$. Let

$$\mathcal{F} := \{t \in \widetilde{T} \mid \varphi_h(t) \otimes \varphi_0(t_0) = \varphi_0(t) \otimes \varphi_h(t_0) \text{ for any } h \in H\}.$$

For each h, the condition is the vanishing of a holomorphic section of $p_T^*M \otimes (p_T^*M)_{t_0}$, so $\mathcal{F}$ is a closed analytic subset of $\widetilde{T}$ [GR84, Chapter 5, §6.1, Corollary of the Proposition].

We show that $\mathcal{F}$ is compact, by an argument of [Gro91]. Let $(t_n)_{n\geq 1}$ be a sequence in $\mathcal{F}$. Since T is compact, there exists a compact subset $K \subset \widetilde{T}$ with $\Gamma K = \widetilde{T}$. Write $t_n = \gamma_n k_n$ with $\gamma_n \in \Gamma$ and $k_n \in K$. Possibly passing to a subsequence, we may assume that k_n converges to a point $k \in K$. Suppose that the set $\{\gamma_n \mid n \geq 1\}$ is infinite. Possibly passing to a subsequence, we may assume that the elements γ_n are pairwise distinct. Since Γ acts properly discontinuously on $\widetilde{T}$, t_n leaves any compact subset of $\widetilde{T}$ for n large, so $|\varphi_0(t_n)| \to 0$ by Construction-Proposition 3.5(4). Let $h \in H$, and let $h_n := (\gamma_n^{-1})^* h \in H$. By Construction-Proposition 3.4(2), $\varphi_{h_n} = (\gamma_n^{-1})^* \varphi_h$, so $|\varphi_{h_n}(t_n)| = |\varphi_h(k_n)|$. Since $t_n \in \mathcal{F}$, Construction-Proposition 3.5(3) implies

$$|\varphi_h(k_n)| \cdot |\varphi_0(t_0)| = |\varphi_0(t_n)| \cdot |\varphi_{h_n}(t_0)| \leq |\varphi_0(t_n)| \cdot \|h\| \to 0.$$

Since $\varphi_0(t_0) \neq 0$, we get $\varphi_h(k) = 0$ for any $h \in H$, which contradicts Construction-Proposition 3.4(3). Thus the set $\{\gamma_n \mid n \geq 1\}$ is finite, so a subsequence of (t_n) converges, and the limit belongs to $\mathcal{F}$ since $\mathcal{F}$ is closed. Hence $\mathcal{F}$ is compact.

Since $p(y_0) \in S^\circ$, $\tilde{\mu}$ is an isomorphism near t_0. Since $r_H(y_0) = r(H) = 1$, there exists $h_1 \in H$ such that $u := h_1/h_0$ satisfies $du(y_0) \neq 0$. Since $r_H \leq r(H) = 1$ at the points where $h_0 \neq 0$, and $du \neq 0$ near y_0, we have $d(h/h_0) \wedge du = 0$ near y_0 for any $h \in H$. Let Σ be the germ at y_0 of the smooth curve $\{u = u(y_0)\}$. Then h/h_0 is constant on Σ for any $h \in H$. For t near t_0 with $\tilde{\mu}(t) \in \Sigma$, we get

$$\frac{\varphi_h(t)}{\varphi_0(t)} = \frac{h}{h_0}(\tilde{\mu}(t)) = \frac{h}{h_0}(y_0) = \frac{\varphi_h(t_0)}{\varphi_0(t_0)}$$

for any $h \in H$, so $t \in \mathcal{F}$. Thus the connected component $\mathcal{F}_0$ of $\mathcal{F}$ containing t_0 is a compact connected analytic subset of $\widetilde{T}$ of positive dimension. By the proper mapping theorem [GR84, Chapter 10, §6.1], $\tilde{\mu}(\mathcal{F}_0)$ is a compact connected analytic subset of $\widetilde{S}$ containing y_0, and $\tilde{\mu}(\mathcal{F}_0)$ has positive dimension since $\tilde{\mu}$ is an isomorphism near t_0. This contradicts Lemma 3.1. Thus $r(H) = 2$. □

Proposition 3.7. *Notation as in Set-up 3.2. Then the elements of H have no common zero on $\widetilde{S}$.*

Proof. Let I be as in Construction-Proposition 3.3, let μ, T, D, and M be as in Construction-Proposition 3.4, and let S°, Ω, and θ be as in Construction-Proposition 3.5. By Lemma 3.6, Ω is open and non-empty, so $p(\Omega) \cap S^\circ$ is open and non-empty. By Construction-Proposition 3.5(1) and (2), θ is semi-positive on T and positive definite on the non-empty open subset $\mu^{-1}(p(\Omega) \cap S^\circ)$ of T.

Since θ represents $c_1(M)$ in $H^2(T,\mathbb{R})$ [Voi02, Theorem 7.10], for any irreducible curve $C \subset T$ with normalization $\nu\colon C^\nu \to C$, we have

$$M \cdot C = \deg \nu^* M = \int_{C^\nu} \nu^*\theta \geq 0,$$

and $M^2 = \int_T \theta \wedge \theta > 0$ [BHPV04, Chapter II, Section 10]. Thus M is nef, and $M^2 \geq 1$ since M^2 is an integer.

Let $A := \mu^* K_S$. By Lemma 2.1(1), A is nef and $A^2 = K_S^2 = 1$. Since M and A are nef and $D \geq 0$, we have

$$1 \leq M^2 = M \cdot A - M \cdot D \leq M \cdot A = A^2 - A \cdot D \leq 1.$$

Thus $M \cdot D = A \cdot D = 0$, so $D^2 = A \cdot D - M \cdot D = 0$. The intersection form on divisors modulo numerical equivalence on T has signature $(1, n-1)$, where n is the Picard number of T [Har77, Theorem V.1.9 and Remark V.1.9.1]. Since $A^2 > 0$, the form is negative definite on the divisors orthogonal to A, so D is numerically trivial. Since $D \geq 0$, we get $D = 0$. Thus $I \cdot \mathcal{O}_T = \mathcal{O}_T$. For any $t \in T$, some $f \in I_{\mu(t)}$ satisfies $f(\mu(t)) \neq 0$, so $\mu(t) \notin V(I)$. Since μ is surjective, $V(I) = \emptyset$, and Construction-Proposition 3.3(4) implies that the elements of H have no common zero on $\widetilde{S}$. □

4. Proof of Theorem 1.1

In this section, we prove Theorem 1.1.

Lemma 4.1. *Notation and conditions as in Set-up 2.3. Assume that S is a numerical Godeaux surface and Γ is infinite. Then $\dim_\Gamma B \geq 1$.*

Proof. By Serre duality and Lemma 2.1(1), $\chi(S, K_S) = \chi(\mathcal{O}_S) = 1$. By Theorem 2.9,

$$1 = \dim_\Gamma B - \dim_\Gamma \mathcal{H}^{2,1}_{(2)}(\widetilde{S}) + \dim_\Gamma \mathcal{H}^{2,2}_{(2)}(\widetilde{S}).$$

The Hodge star operator $*$ of $p^*\omega_S$ is an isometry, and $*$ commutes with the Laplacian Δ_d of d [Voi02, p. 134]. Since $p^*\omega_S$ is Kähler, $\Delta_d = 2\Delta_{\bar\partial}$ [Voi02, Theorem 6.7]. Thus $*$ maps $\mathcal{H}^{2,2}_{(2)}(\widetilde{S})$ isomorphically onto $\mathcal{H}^{0,0}_{(2)}(\widetilde{S})$, which is $H^0_{(2)}(\widetilde{S}, \mathcal{O}_{\widetilde{S}})$ by Theorem 2.9.

Let $u \in H^0_{(2)}(\widetilde{S}, \mathcal{O}_{\widetilde{S}})$. By Lemma 2.6(2), applied to the trivial line bundle with the trivial metric, $|u|$ tends to zero at infinity. Thus $|u|$ attains a maximum on $\widetilde{S}$, and u is constant by the maximum principle. Since Γ is infinite, $\widetilde{S}$ has infinite volume, so $u = 0$. Hence $\mathcal{H}^{2,2}_{(2)}(\widetilde{S}) = 0$, and

$$\dim_\Gamma B = 1 + \dim_\Gamma \mathcal{H}^{2,1}_{(2)}(\widetilde{S}) \geq 1.$$

□

Set-up 4.2. Notation and conditions as in Set-up 2.3. Let x, s, e_s, $\ell^2(\Gamma)$, and E_x be as follows.

(1) S is a numerical Godeaux surface.
(2) Γ is infinite.
(3) $x \in \widetilde{S}$ is a point.
(4) $s := p(x)$.
(5) e_s is a non-zero element of the fiber $(K_S)_s$ of K_S at s.
(6) $\ell^2(\Gamma)$ is the Hilbert space of families $c = (c_\gamma)_{\gamma\in\Gamma}$ of elements of $\mathbb{C}$ with $\sum_\gamma |c_\gamma|^2 < +\infty$, with the norm $\|c\| = (\sum_\gamma |c_\gamma|^2)^{1/2}$.
(7) For $h \in B$, $E_x(h) := \big(h(\gamma x)/e_s\big)_{\gamma\in\Gamma}$, where $h(\gamma x)$ belongs to the fiber of p^*K_S at γx, which is $(K_S)_s$.

Lemma 4.3. *Notation as in Set-up 4.2. Then E_x is an injective bounded linear operator from B to $\ell^2(\Gamma)$, and the image of E_x is dense in $\ell^2(\Gamma)$.*

Proof. By Lemma 2.6(1), E_x is a bounded linear operator from B to $\ell^2(\Gamma)$. For $\delta \in \Gamma$ and $c = (c_\gamma)_{\gamma\in\Gamma} \in \ell^2(\Gamma)$, let $c^\delta := (c_{\delta\gamma})_{\gamma\in\Gamma}$. Since $(\delta^* h)(\gamma x) = h(\delta\gamma x)$, we have

$$E_x(\delta^* h) = E_x(h)^\delta \tag{4.1}$$

for any $h \in B$ and $\delta \in \Gamma$. Thus $\ker E_x$ is a closed Γ-invariant subspace of B, and any element of $\ker E_x$ vanishes at x. By Proposition 3.7, applied to $H = \ker E_x$ if $\ker E_x \neq 0$, we get $\ker E_x = 0$.

Let $\phi \in L^2(\widetilde{S}, K_{\widetilde{S}})$ with $\|\phi\| = 1$ and $\phi = 0$ outside $\widetilde{S}_0$. For $\gamma \in \Gamma$, the section $(\gamma^{-1})^*\phi$ vanishes outside $\gamma\widetilde{S}_0$, so the sections $(\gamma^{-1})^*\phi$ for $\gamma \in \Gamma$ form an orthonormal family. Let

$$\iota\colon \ell^2(\Gamma) \to L^2(\widetilde{S}, K_{\widetilde{S}}), \qquad c \to \sum_{\gamma\in\Gamma} c_\gamma (\gamma^{-1})^*\phi.$$

Then ι is an isometry, and $\delta^*\iota(c) = \iota(c^\delta)$ for any $\delta \in \Gamma$. Thus $V_\phi := \iota(\ell^2(\Gamma))$ is a closed Γ-invariant subspace of $L^2(\widetilde{S}, K_{\widetilde{S}})$. Since $(\gamma^{-1})^*\phi = 0$ on $\widetilde{S}_0$ for any $\gamma \neq 1$, Definition 2.7 implies

$$\dim_\Gamma V_\phi = \sum_{\gamma\in\Gamma} \int_{\widetilde{S}_0} |(\gamma^{-1})^*\phi|^2 dV = \int_{\widetilde{S}_0} |\phi|^2 dV = 1.$$

Let P_B be the orthogonal projection of $L^2(\widetilde{S}, K_{\widetilde{S}})$ onto B, and let $\Psi := \iota \circ E_x \circ P_B$. Since B is Γ-invariant and the operators δ^* are unitary, P_B commutes with δ^*, so $\Psi\delta^* = \delta^*\Psi$ for any $\delta \in \Gamma$ by (4.1). Since $\ker E_x = 0$, we have $\ker \Psi = B^\perp$, so $(\ker \Psi)^\perp = B$. Since ι is an isometry, the closure of the image of Ψ is $\iota(\overline{E_x(B)}) \subset V_\phi$. By Lemma 2.8(2) and (1),

$$\dim_\Gamma B = \dim_\Gamma \iota(\overline{E_x(B)}) \leq \dim_\Gamma V_\phi = 1.$$

By Lemma 4.1, $\dim_\Gamma B \geq 1$, so $\dim_\Gamma \iota(\overline{E_x(B)}) = \dim_\Gamma V_\phi = 1$. By Lemma 2.8(1), $\iota(\overline{E_x(B)}) = V_\phi$, so $\overline{E_x(B)} = \ell^2(\Gamma)$. □

Proof of Theorem 1.1. Let S be a numerical Godeaux surface, and suppose that $\Gamma := \pi_1(S)$ is infinite. Let x, s, e_s, $\ell^2(\Gamma)$, and E_x be as in Set-up 4.2. Let

$$Q\colon \ell^2(\Gamma) \to \mathbb{C}, \qquad c \to \sum_{\gamma\in\Gamma} c_\gamma^2.$$

By the Cauchy–Schwarz inequality, for any $c, c' \in \ell^2(\Gamma)$,

$$|Q(c) - Q(c')| \leq \|c - c'\| \cdot (\|c\| + \|c'\|),$$

so Q is continuous. Since Q takes the value 1 at the element of $\ell^2(\Gamma)$ which is 1 at the identity and 0 elsewhere, and since $E_x(B)$ is dense in $\ell^2(\Gamma)$ by Lemma 4.3, there exists $h \in B$ with $Q(E_x(h)) \neq 0$.

The section $h^2 := h \otimes h$ of $p^*(2K_S)$ is integrable, since $\int_{\widetilde{S}} |h^2| dV = \|h\|^2$ [Kol95, (5.12.3)], so by [Kol95, (5.14.1)], the series $\sum_{\gamma\in\Gamma} |h^2(\gamma y)|$ converges uniformly for y in compact subsets of $\widetilde{S}$. Thus the Poincaré series $\sum_{\gamma\in\Gamma} \gamma^*(h^2)$ [Kol95, Definition 5.10] converges uniformly on compact subsets of $\widetilde{S}$, and the sum of the Poincaré series is a Γ-invariant holomorphic section of $p^*(2K_S)$. Hence the sum is $p^*\sigma$ for a holomorphic section σ of $2K_S$, which belongs to $H^0(S, 2K_S)$ by GAGA [Har77, Appendix B, Theorem 2.1], and

$$\sigma(s) = \sum_{\gamma\in\Gamma} h(\gamma x)^{\otimes 2} = Q(E_x(h)) \cdot e_s^{\otimes 2} \neq 0.$$

Thus s is not a base point of $|2K_S|$. Since $x \in \widetilde{S}$ is arbitrary, $|2K_S|$ has no base point, which contradicts Lemma 2.1(2). Therefore Γ is finite.

Since Γ is finite, Γ is isomorphic to the profinite completion of Γ, so Γ is a cyclic group of order at most 5 by Theorem 2.2. Since Γ is abelian, $\Gamma \cong H_1(S, \mathbb{Z})$ by the Hurewicz theorem [Hat02, Theorem 2A.1], so $H_1(S, \mathbb{Z})$ is finite and equal to the torsion subgroup of $H_1(S, \mathbb{Z})$. By

Lemma 2.1(3), Γ is isomorphic to the torsion subgroup of $\mathrm{Pic}(S)$. In particular, S is simply connected if and only if the torsion subgroup of $\mathrm{Pic}(S)$ is trivial. □

References

[Bar85] R. Barlow, *A simply connected surface of general type with $p_g = 0$*, Invent. Math. **79** (1985), no. 2, 293–301.

[BHPV04] W. Barth, K. Hulek, C. Peters, and A. Van de Ven, *Compact complex surfaces*, 2nd ed., Ergeb. Math. Grenzgeb. (3) **4**, Springer-Verlag, Berlin (2004).

[BCP11] I. Bauer, F. Catanese, and R. Pignatelli, *Surfaces of general type with geometric genus zero: a survey*, in: *Complex and differential geometry*, Springer Proc. Math. **8**, Springer, Heidelberg (2011), pp. 1–48.

[Cam94] F. Campana, *Remarques sur le revêtement universel des variétés kählériennes compactes*, Bull. Soc. Math. France **122** (1994), no. 2, 255–284.

[CCO94] F. Catanese, P. Cragnolini, and P. Oliverio, *Surfaces with $K^2 = \chi = 2$ and special nets of quadrics in 3-space*, in: *Classification of algebraic varieties (L'Aquila, 1992)*, Contemp. Math. **162**, Amer. Math. Soc., Providence, RI (1994), pp. 77–128.

[CD89] F. Catanese and O. Debarre, *Surfaces with $K^2 = 2$, $p_g = 1$, $q = 0$*, J. Reine Angew. Math. **395** (1989), 1–55.

[CU18] S. Coughlan and G. Urzúa, *On $\mathbb{Z}/3$-Godeaux surfaces*, Int. Math. Res. Not. IMRN (2018), no. 18, 5609–5637.

[CG94] P. Craighero and R. Gattazzo, *Quintic surfaces of $\mathbb{P}^3$ having a non singular model with $q = p_g = 0$, $P_2 \neq 0$*, Rend. Sem. Mat. Univ. Padova **91** (1994), 185–198.

[DR26] E. Dias and C. Rito, *$\mathbb{Z}/2$-Godeaux surfaces*, J. Algebra **701** (2026), 340–357.

[FR18] M. Franciosi and S. Rollenske, *Canonical rings of Gorenstein stable Godeaux surfaces*, Boll. Unione Mat. Ital. **11** (2018), no. 1, 75–91.

[God31] L. Godeaux, *Sur une surface algébrique de genres zéro et de bigenre deux*, Atti Accad. Naz. Lincei Rend. Cl. Sci. Fis. Mat. Natur. (6) **14** (1931), 479–481.

[GR84] H. Grauert and R. Remmert, *Coherent analytic sheaves*, Grundlehren Math. Wiss. **265**, Springer-Verlag, Berlin (1984).

[Gro91] M. Gromov, *Kähler hyperbolicity and L_2-Hodge theory*, J. Differential Geom. **33** (1991), no. 1, 263–292.

[Har77] R. Hartshorne, *Algebraic geometry*, Grad. Texts in Math. **52**, Springer-Verlag, New York–Heidelberg (1977).

[Hat02] A. Hatcher, *Algebraic topology*, Cambridge Univ. Press, Cambridge (2002).

[Kol95] J. Kollár, *Shafarevich maps and automorphic forms*, M. B. Porter Lectures, Princeton Univ. Press, Princeton, NJ (1995).

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. II*, Ergeb. Math. Grenzgeb. (3) **49**, Springer-Verlag, Berlin (2004).

[LLZ26] J. Liu, W. Liu, and L. Zhang, *Classification of numerical Godeaux surfaces*, preprint (2026), https://jihaoliu.org/papers/classification-of-numerical-godeaux-surfaces.html.

[Miy76] Y. Miyaoka, *Tricanonical maps of numerical Godeaux surfaces*, Invent. Math. **34** (1976), no. 2, 99–111.

[RTU17] J. Rana, J. Tevelev, and G. Urzúa, *The Craighero–Gattazzo surface is simply connected*, Compos. Math. **153** (2017), no. 3, 557–585.

[Rei] M. Reid, *Godeaux and Campedelli surfaces*, notes, https://mreid.warwick.ac.uk/surf/more/Godeaux.pdf.

[Rei78] M. Reid, *Surfaces with $p_g = 0$, $K^2 = 1$*, J. Fac. Sci. Univ. Tokyo Sect. IA Math. **25** (1978), no. 1, 75–92.

[Voi02] C. Voisin, *Hodge theory and complex algebraic geometry. I*, Cambridge Stud. Adv. Math. **76**, Cambridge Univ. Press, Cambridge (2002).

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China
Email address: liujihao@math.pku.edu.cn

School of Mathematical Sciences, Xiamen University, Xiamen, Fujian 361005, China
Email address: wliu@xmu.edu.cn

School of Mathematical Sciences, University of Science and Technology of China, 96 Jinzhai Road, Baohe District, Hefei 230026, China
Email address: zhlei18@ustc.edu.cn